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| Student's Name | .....    | Grade     | 10 |
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### **Q1: choose the correct answer:**

#### **1-What is the primary goal of supervised learning?**

- a) To discover hidden patterns in unlabeled data.
- b) To predict an output variable based on input features using labeled data. ☐
- c) To reduce the dimensionality of data.
- d) To optimize a reward function through trial and error.

#### **2-Which of the following is an example of a classification algorithm?**

- a) Linear Regression
- b) K-Means Clustering
- c) Decision Trees ☐
- d) Principal Component Analysis (PCA)

#### **3-What does the term "overfitting" mean in ML?**

- a) The model performs well on training data but poorly on unseen data. ☐
- b) The model is too simple to capture patterns in the data.
- c) The model has high bias.
- d) The model ignores noise in the data.

#### **4-Which evaluation metric is most suitable for imbalanced classification problems?**

- a) Accuracy
- b) Mean Squared Error (MSE)
- c) F1-Score ☐
- d) R-Squared



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### Q1: Compare and contrast decision trees and logistic regression:

- **Decision Trees:**

- Non-linear decision boundaries
- Can handle mixed feature types
- Interpretable as rules
- Prone to overfitting
- No need for feature scaling

- **Logistic Regression:**

- Linear decision boundary (without transformations)
- Requires numerical features
- Probabilistic interpretation
- Less prone to overfitting with regularization
- Requires feature scaling for best performance

### Short Answer Questions:

\*Why do we split data into training and test sets?

**Answer:** To evaluate how well the model generalizes to unseen data and detect overfitting.



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**Q1 : Read and put True( ✓) or false (✗) :**

- 1- **Supervised learning** requires labeled data for training. **(True)**
- 2- **Unsupervised learning** deals with unlabeled data. **(True)**
- 3- **Overfitting** occurs when a model performs well on training data but poorly on unseen data. **(True)**
- 4- **Classification** is a type of regression problem. **(False)**
- 5- **Linear regression** is used for predicting continuous values. **(True)**
- 6- **Decision trees** can only be used for classification tasks. **(False)**
- 7- **K-means clustering** is a supervised learning algorithm. **(False)**
- 8- **Feature scaling** is essential for algorithms like SVM and KNN. **(True)**
- 9- **Deep learning** is a subset of reinforcement learning. **(False)**
- 10- **Bias** in a model refers to the error introduced by approximating real-world problems. **(True)**
- 11- **Cross-validation** helps in reducing overfitting. **(True)**
- 12- **PCA (Principal Component Analysis)** is used for feature extraction. **(True)**
- 13- **Reinforcement learning** relies solely on historical data without any interaction. **(False)**
- 14- **A confusion matrix** is used to evaluate classification models. **(True)**
- 15- **Gradient descent** is used to minimize the loss function. **(True)**